

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (90 minutes) April 6th, 2013

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. A random sample of 20 adult Colobus monkeys is obtained from a jungle in Angola. The following stem & leaf gives the results of a Monkey IQ test performed on these animals with test scores ranging from 0 = least intelligent to 50 = most intelligent

1 | 1,4,8,9
2 | 0,2,3,4,5,6,7,8
3 | 1,1,3,5,8
4 | 0,3,6

- Compute the median Monkey-IQ in the sample_____
- Compute the interquartile range of Monkey-IQ in the sample_____
- It is known that the Median Monkey-IQ of an adult Chacma Baboon is 37. Perform a hypothesis test of the null hypothesis that the median Monkey-IQ of an adult Colobus monkey is the same as this against the alternative that it is not at $\alpha = 0.01$
- A scientist believes that 10% of these Colobus Monkeys have a monkey IQ of 47 or greater. Is this refuted by the data, explain your answer?

2. The Heights of Asperella cacti at 1 year from being planted are $N(\mu=80, \sigma^2=900)$. A botanist has developed a fertilizer that he believes will increase the mean height to 86 but not change the standard deviation $N(\mu=86, \sigma^2=900)$. He will test this belief on the increase in height as follows

- n cacti without fertilizer will be planted (Group A)
 - The same number n with fertilizer will be planted (Group B)
 - The heights in Groups A and B at one year will be compared at a two sided $\alpha = 0.05$ using the best test to make this comparison
- What is the best statistical approach to test for the null hypothesis (i.e the name of the test that would be used) and what assumptions, if any, in addition to those already given above are needed
 - Give a large sample approximation estimate of the smallest number of Asperella that must be planted for which there is 90% power to reject H_0 if the botanist's belief on the fertilizer is correct
 - Suppose now that the botanist had chosen $n = 1000$ and conducted the experiment, but that for $\frac{1}{2}$ of the plants in Group B the fertilizer was misapplied so that it could have no effect on Height. Estimate the power to reject H_0 in this setting if the botanist's belief on the fertilizer (when correctly applied) is correct.

3. A Statistics Professor believes that adding videos of statisticians consulting on applied problems to her standard instructional material will help students solve problems. She designs an experiment to find evidence to support this hypothesis. A total of 167 students are given a test prior to receiving her class (**the “pretest”**) to see how they solve applied statistics problems. The pretest is graded Pass or Fail. The students are then randomly divided into two groups: The Experimental Group with 106 students, and the Control Group with 61 students. Over the next 2 weeks, the Experimental group are shown videos plus her standard instructional material, while the Control group receives only her standard instructional material. After this is done, all students take the same Pass/Fail test again on solving applied statistics problems (the “post-test”). You can assume that all students are independent. The following are the combined Pretest Posttest outcomes:

Group	Pretest	Posttest	Number of subjects
Experimental	Fail	Fail	37
Experimental	Fail	Pass	29
Experimental	Pass	Fail	11
Experimental	Pass	Pass	29
Control	Fail	Fail	26
Control	Fail	Pass	9
Control	Pass	Fail	7
Control	Pass	Pass	19

- Using only the data for the experimental group is there statistically significant evidence of an improvement between Pretest and Posttest performance.? What is the Null Hypothesis; the Alternative Hypothesis, and show the details for the test used ?
- Using only the data for the control group is there statistically significant evidence of an improvement between the pretest and posttest?
- Now combine the analyses for the experimental and control groups to see if there is statistically significant evidence that adding the video to the instruction was helpful.

4. A large fertilizer Company seeks to design an experiment to study the effect of minerals in fertilizer mixes used to grow corn. Two levels (low, high) are used for mineral A, and two levels (low, high) are used for mineral B. Three varieties of corn are included in the study. The response variable is yield per acre.

A sample of 6 farms are picked at random from the farms in the State to participate in the study. Each farm will have 12 randomly picked plots to be planted in corn for use in the experiment. There are thus a total of 72 plots in the experiment.

You are asked to consider two alternative designs for the experiment in a) and b) below. For each design, write down the data layout, the ANOVA table with sources of variability, degrees of freedom, Expected Means Squares, and indicate which F-ratios to take. Include in your model all estimable interactions (including interaction with Farms). Explain based on your analysis the pros and cons of the two designs.

- Design I: For Farm 1 assign completely at random the 12 plots on the farm to the 12 combinations of levels of A,B, and Variety, so that each combination will appear exactly once on Farm 1. Do a similar but independent randomization for each of the six farms.
- Design II: Pick at random two of the farms for variety 1, two farms for variety 2, and two farms for variety 3. For each farm assign at random 3 plots to each of the combinations of levels of Factors A and B.